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Automatic methods for finding similar press news articles and comparing their features

Requirements specification

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1 Abstract

This document contains a describtion of the thesis: Automatic methods for finding similar press news articles and comparing their features. It describes main goal and vision of how the work will be done, presents functional and non-functional requirements, analyses potiential risks and shows schedule of work in project.

2 Vocabulary

NLP - Natural Language Processing, it combines computational linguistics—rule-based modeling of human language with statistical, machine learning, and deep learning models.

Semantic Textual Similarity - the task of evaluating how similar two texts are in terms of meaning.

Visualization - Presenting results of the analysis with numbers, graphs or text differences.

3 Specification

3.1 Executive Summary

The objective of this project is to create a dataset and develop techniques for identifying similar news articles along with their distinctive characteristics. This involves training machine learning models and creating approaches for extracting textual and structural attributes that are used in a similarity metric designed to evaluate the resemblance between news articles. User should see basic graphic interface which allow him to upload files to compare and see the result. Furthermore, the project includes the creation of smart visual representations and explanations for the identified similarities.

Up to this point, available datasets and solutions for NLP tasks have primarily been tailored for shorter text and encompass more general subject matter. The primary objective of this thesis is to formulate methodologies for comparing longer documents. This necessitates the development of text features and techniques to construct representations for these diverse and extensive documents, which may originate from sources such as the News API or a proprietary database provided by an industry collaborator.

3.2 Functional Requirements

Asking what can we expect from our tool, there is Table 1 which presents us functional requirements of it. We can find only one actor who is actually a user of the tool, the name of action and short description what the user can do and how the tool will behave. Below table we can find user stories.

Actor	Name	Description
User	Text input	The tool should allow users to input or upload text documents for comparison. It should support various text formats such as plain text, PDF, Word documents, or URLs.
	Customization	Allow users to select and configure the similarity algorithms and parameters, enabling them to tailor the comparison to their specific needs.
	Visualization	Offer visual representations of the compared texts, such as side-by-side comparisons, highlighting differences, or showing similarity scores graphically.
	Comparing articles	User can compare two or more articlesin terms of content and text

Table 1: Functional requirements

1. As a user I want to open file dialog so I can choose file from my device.
2. As a user I want to paste an url so I can choose file from internet to analyse.
3. As a user I want to paste plain text so I can analyse it.
4. As a user I want to choose algorithm of comparsion so I can choose analysing method.
5. As a user I want to see result as chosen metric so I will know if texts are similar.
6. As a user I want to see visualization of comparsion so I will know which parts of texts are similar.

3.3 Non-Functional Requirements

In table 2 we can find non-functional requirements which focus in the area of usability, reliability, performance and supportability.

Requirements area	No.	Description
Usability	1.	Support multiple languages and regional preferences, allowing users to interact with the tool in their preferred language.
	2.	The tool's interface and workflows should be intuitive, with clear and self-explanatory controls and actions.
	3.	Ensure that the tool works well on various devices, browsers, and operating systems, providing a consistent experience.
Reliability	5.	Inputing incorrect data by the user should not interrupt the operation of the application and should display appropriate error messages.
Performance	6.	The tool should provide fast and responsive results, even when handling large datasets, and should have low latency for text comparison.
Supportability	7.	Developers could easily add new method of comparing by creating a new class inherited from base class.

Table 2: Non-functional requirements

4 Project schedule

Project is intended to start with the beginning of October 2023 and finish in January 2024. Project schedule with most important milestones is presented in Table 3.

Task	Time
Project initiation • Define project objectives, scope, and requirements. • Identify stakeholders and their expectations. • Create a detailed project plan and schedule.	02.10 - 15.10
Research and Data Collection • Conduct a literature review to understand relevant NLP techniques. • Identify and gather the necessary data for the project. • Preprocess and clean the data for analysis. • Explore potential data sources and APIs.	16.10 - 22.10
Design and develop NLP models and algorithms • Build the NLP model • Implement the neural network layers	23.10 - 12.11
Train and fine-tune NLP models using the collected data • Train and fine-tune NLP models using the collected data • Determine the fine-tuning strategy • Monitor training metrics	13.11 = 03.12
Integrate the developed NLP models into the project framework	04.11 - 10.12
Conduct thorough testing, including unit testing and integration testing • Address any bugs, issues, or optimization requirements	11.12 - 24.12
Prepere final report • Document the project's progress, methodologies, and findings. • Generate regular status reports for stakeholders. • Prepare documentation for future reference and knowledge sharing.	25.12 - 12.01

Table 3: Project schedule

5 Risk analysis

Potential risks for the project were analyzed in Table 4 according to the SWOT methodology (Strengths, Weaknesses, Opportunities, Threats). Threats and weaknesses were given a proposals to how can I deal with them.

Strenghts	**Weaknesses**
Knowledge in area of algorithms	Lack of previous experience with NLP
Previous experience in coding in Python and using machine learning algorithms	Time constraints due to professional work and other projects
Opportunities	**Threats**
Interesting topic showing one of problems in real life	Possible difficulties with creating methods for large articles
Many NLP algorithms repositories and well documented Python libraries	Lack of experience in seeking for articles in different languages

Table 4: SWOT Analysis

1. Lack of previous experience with NLP - To close a knowledge gap in NLP I will focus on existing solutions, begin by researching and studying NLP tools and libraries like NLTK, spaCy, and Gensim, which offer pre-built solutions for common NLP tasks. Explore open-source projects and pre-trained models, like BERT or Word2Vec, to leverage existing resources.
2. Time constraints due to professional work and other projects - : Completion of all additional projects, assignments and tests as soon as possible, in order to devote the rest of the semester to the thesis. Determining the time during the work week that I can use for thesis-related work. Real-time control of the schedule progress.
3. Possible difficulties with creating methods for large articles - There are solutions for shorter text analysis. Combining or experimenting with smaller tasks can in coclusion give a satisfying result in bigger task.
4. Lack of experience in seeking for articles in different languages - there are datasets published by foreign provider and NEWS API.

6 Bibliography

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